

I1 extends I2, I3

Program Code:

```
package core_java;

interface Device {
    void powerOn();
}

interface Camera {
    void takePhoto();
}

interface Smartphone extends Device, Camera {
    void browseInternet();
}

class iPhone implements Smartphone {
    @Override
    public void powerOn() {
        System.out.println("iPhone is powering on...");
    }

    @Override
    public void takePhoto() {
        System.out.println("I want to take a photo.");
    }

    @Override
    public void browseInternet() {
        System.out.println("Browsing websites on 5G network...");
    }
}

public class CtoMultipleInterfaces {
    public static void main(String[] args) {
        iPhone myPhone = new iPhone();
        myPhone.powerOn();
        myPhone.takePhoto();
        myPhone.browseInternet();
    }
}
```

Explanation:

This program demonstrates multiple inheritance using interfaces. Smartphone extends both Device and Camera, so it inherits the methods powerOn() and takePhoto() and also declares browseInternet(). The iPhone class implements Smartphone and provides implementations for all three methods.

Compilation and Execution Commands:

Save the file as: CtoMultipleInterfaces.java

Compile:

```
javac -d . CtoMultipleInterfaces.java
```

Run:

```
java core_java.CtoMultipleInterfaces
```

Output Screenshot:

```
iPhone is powering on...

I want to take a photo.

Browsing websites on 5G network...
```

Verification:

The output confirms that the iPhone object successfully executes methods inherited through the Smartphone interface from both Device and Camera, along with its own browseInternet() method.

Result:

Thus, the Java program was successfully executed and multiple inheritance through the interface relationship I1 extends I2, I3 was verified successfully.